Arham Lodha

Los Angeles, CA | arham.lodha@utexas.edu | github.com/arham-lodha | linkedin.com/in/arham-lodha

Education

University of California, Los Angeles Los Angeles, CA
PhD in Mathematics *Aug 2026 — May 2031*

- NSF Graduate Research Fellow

University of Texas at Austin Austin, TX
BS in Mathematics *Aug 2022 — May 2026*

- GPA: 3.9314/4.0000
- **Relevant Coursework:** Algebraic Structures I and II, Algebraic Topology I, Real Analysis I, Complex Analysis, Harmonic Analysis and Representation Theory, Representation Theory, Riemann Surfaces, Honors Thesis Course, Probability I, Numerical Analysis, Honors Computer Graphics
- **Exchange Program at ETH Zurich (Spring 2025):** Introduction to Modular Forms, Algebraic Topology II, Applied Stochastic Processes, Decentralized Finance

Research Experience

Undergraduate Researcher Jan 2026 — May 2026
University of Texas at Austin *Advisor: Professor Lorenzo Sadun*

- Conducted research in extremal combinatorics and graph limit theory, formulating an original problem with guidance from advisor on the asymptotic behavior of homomorphism densities in graphons constrained by fixed edge density ε and vanishing triangle density $\tau \to 0$.
- Established the main result $t_G(\varepsilon, \tau) = \Theta\big(\tau^{\alpha(G)}\big)$, where $\alpha(G)$ is the optimal value of a linear program whose constraints are dictated by the triangle structure of G, precisely quantifying the rate at which higher-order interactions decay when 3-particle interactions are suppressed.
- Preparing manuscript for publication; currently extending results to broader domains and sharper constants.

DIMACS REU Participant May 2025 — Jul 2025
Rutgers University *Principal Investigator: Professor Feng Luo*

- Conducted research in discrete conformal geometry, focusing on circle packings, to investigate two core problems: the Rigidity of Circle Packings and the Discrete Schwarz-Pick Lemma.
- Proved the Discrete Schwarz-Pick Lemma for a broader class of inversive distances, extending prior results limited to tangent cases. Established that the space of admissible radii remains convex and curvature remains monotonic under these weaker conditions.
- Designed Python experiments to systematically test boundary configurations; utilized these simulations to construct a counterexample that disproved the conjecture for disjoint circle packings.
- Presented findings at the DIMACS REU End-of-Program Presentations. Authored a manuscript summarizing these results (available on arXiv); Submitted manuscript for publication.

Research Fellow May 2023 — Jan 2025
University of Texas at Austin *Mentor: Dr. Wenrui Chai*

- Researched the performance of global optimization algorithms that find global minimum potential energy configurations of complex chemical systems. Evaluated and compared the performance of Basin Hopping and Minimum Hopping optimization algorithms and various hybrids.

Independent Study

UT Directed Reading Program — Fourier Analysis Jun 2024 — Jul 2024

- Completed intensive study of Fourier Analysis through "Fourier Series and Integrals" by H. Dym and H.P. McKean, mentored by Ph.D. student William Winston.
- Presented original talk connecting Fourier Analysis to the Heisenberg-Weyl Uncertainty Principle at the UT Directed Reading Program Symposium.

UT Directed Reading Program — Matroid Theory Sep 2023 — Nov 2023

- Studied Matroid Theory under Ph.D. student Jayden Wang using “Matroid Theory: A Geometric Perspective” by Gary Gordon.
- Applied matroid theory to solve a combinatorial problem: determined minimum verification checks for Sudoku puzzles.
- Presented findings at the UT Directed Reading Program Symposium, demonstrating matroid applications in Sudoku verification.

Teaching Experience

Undergraduate Course Assistant Aug 2024 — May 2026
University of Texas at Austin
- Introduction to Mathematics (Jan 2026 – May 2026)
- Business Calculus (Aug 2025 – Dec 2025)
- Probability I (Aug 2024 – Dec 2024)

Grader Jan 2026 — May 2026
University of Texas at Austin
- Discrete Mathematics (Jan 2026 – May 2026)

Leadership Experience

Cofounder Aug 2024 — Present
Hit Fantasy
- Led a cross-functional team of 5 developers and designers to build and launch a bootstrapped fantasy cricket application from concept to deployment. Architected and oversaw a full-stack solution leveraging Figma for UI/UX design and a modern tech stack (Firebase, Next.js, TypeScript).
- Grew active user base from 1,000 users during the 2025 IPL season to 2,000 users during the 2026 IPL season.

Papers & Preprints

- **Lodha, Arham Rajendra.** “The Discrete Schwarz-Pick Lemma For Circle Packings Revisited.” *arXiv preprint* arXiv:2511.10703 (2025). (Submitted for Publication)
- **Lodha, Arham Rajendra.** “Asymptotic Bounds on Homomorphism Densities in Triangle-Constrained Graphons.” *Honors Thesis*, University of Texas at Austin (2026).

Projects

Autograd github.com/arham-lodha/autograd
- Built a from-scratch automatic differentiation library implemented twice: as a NumPy prototype in Python and as a compiled C++ library, both sharing a common three-phase pipeline of symbolic graph construction, an optimizing compiler, and a separate executor.
- Compiler runs multi-pass optimizations over a working copy and emits a flat, topologically-sorted program without mutating the original graph; symbolic differentiation appends gradient nodes via reverse-mode chain rule traversal.
- C++ implementation adds a custom tensor type and a persistent execution buffer to avoid re-allocating storage across forward passes.

Black Hole Raytracer github.com/arham-lodha/Black-Hole-Renderer
- Inspired by *Interstellar*, developed a high-fidelity black hole renderer in Unity using Compute Shaders and advanced ray marching techniques.
- Simulated relativistic effects with precision by integrating the Schwarzschild metric and dynamically adjusting ray trajectories through a modified 4th-order Runge-Kutta integrator.
- Designed and rendered an artistic visualization of a black hole’s accretion disk, capturing phenomena like Black Body Radiation and astrophysical properties.

Skeletal Animation Engine github.com/arham-lodha/Mannequin
- Built a real-time skeletal animation system with interactive bone picking via ray-cylinder intersection, quaternion-based bone manipulation with drag and roll, and linear blend skinning on the GPU.
- Implemented keyframe animation with multiple interpolation modes (linear, cubic spline, circular, bounce easing), arbitrary durations, reorderable keyframes, and camera path interpolation via SLERP.

- Designed an interactive timeline UI with HTML Canvas featuring easing curve previews, per-keyframe duration controls, a rendered keyframe preview sidebar with scrolling, and real-time animation playback.

Minecraft github.com/arham-lodha/Minecraft

- Built a voxel engine with procedural terrain using multi-octave value noise, seamless chunk stitching, lazy loading, and instanced rendering with height-based procedural textures via 3D Perlin noise.
- Developed time-varying animated textures, a multi-level LOD system, a real-time day/night cycle with dynamic lighting, and FPS controls with collision detection and gravity.

Raytracer github.com/arham-lodha/RayTracer

- Built a Whitted-style raytracer with Phong shading, recursive reflection/refraction, shadow attenuation, texture mapping, and Constructive Solid Geometry (union, intersection, set-minus) with interval-based ray classification.
- Optimized with a two-level BVH using surface area heuristic, adaptive supersampling antialiasing, and OpenMP parallelization.

Pokémon DeFi github.com/arham-lodha/PokemonDefi

- Built a decentralized marketplace for trading Pokémon cards as ERC721 NFTs on Ethereum, supporting fixed-price listings and auctions with anti-sniping bid extension and a withdrawal pattern for reentrancy safety.
- Smart contracts written in Solidity using OpenZeppelin (ERC721Enumerable, AccessControl, ReentrancyGuard, Pausable); frontend built with Next.js, Viem, and Wagmi using a simulate-then-write pattern for precise contract error reporting.

Honors and Awards

- **NSF Graduate Research Fellowship**, National Science Foundation (2026)
- **Eva Stevenson Woods Endowed Presidential Scholarship**, University of Texas at Austin (June 2025)
- **William F. Massey Scholarship Fund**, University of Texas at Austin (June 2025)
- **University Honors**, University of Texas at Austin (Fall 2022 – Fall 2025)

Technical Skills

- **Proficient:** Python, Julia, C++, Java, GLSL, HLSL, C#, JavaScript, TypeScript, Dart, Solidity
- **Basic:** MATLAB